



Department of Computer Science and Engineering

Semester: Fall 2023

Course Code: CSE 207

Course Title: Data Structures

Lab: 01

Submitted to

Md. Manowarul Islam

Adjunct Faculty, Department of CSE

Submitted by

Swarna Rani Dey

ID: 2022-1-60-340

Section: 7

Exercise -1

Code:

```
#include <stdio.h>

void bubbleSort(int a[], int n, int *passCount, int *swapCount) {
    int i, j, temp;

    for (i = 0; i < n - 1; i++) {
        (*passCount)++;
        for (j = 0; j < n - 1 - i; j++) {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
                (*swapCount)++;
            }
        }
    }
}

int main() {
    int a[100];
    int n, i, passCount = 0, swapCount = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);
```

```

printf("Enter your data:\n");
for (i = 0; i < n; i++)
    scanf("%d", &a[i]);

printf("Your array: ");
for (i = 0; i < n; i++)
    printf("%d\t", a[i]);

bubbleSort(a, n, &passCount, &swapCount);

printf("\nYour sorted array: ");
for (i = 0; i < n; i++)
    printf("%d\t", a[i]);

printf("\nNumber of passes: %d\n", passCount);
printf("Number of swaps: %d\n", swapCount);

return 0;
}

```

Output:

```

Enter number of elements: 5
Enter your data:
22
22
34
67
56
Your array: 22  22  34  67  56
Your sorted array: 22  22  34  56  67
Number of passes: 4
Number of swaps: 1
...Program finished with exit code 0
Press ENTER to exit console.

```

Exercise -2

Code:

```
#include <stdio.h>

int main() {
    int x[50], n, i, j;

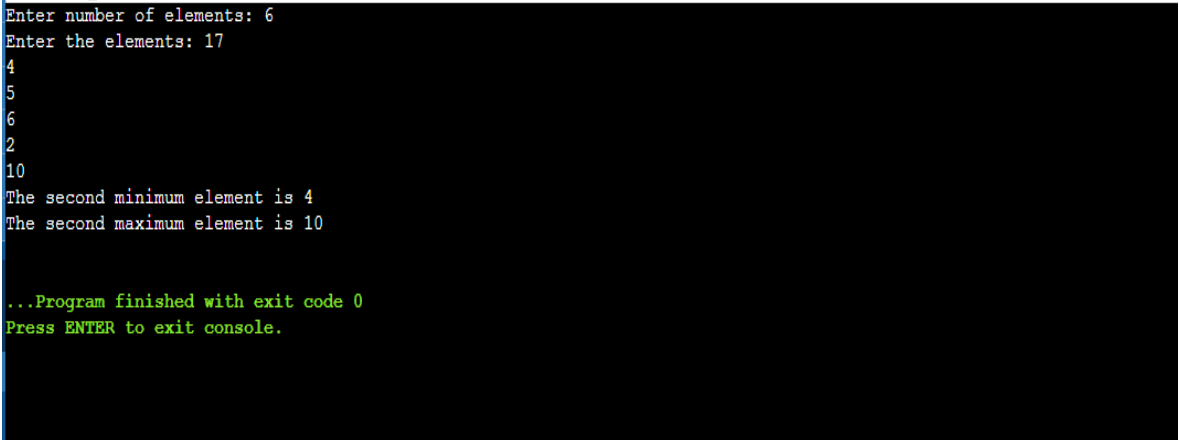
    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &x[i]);
    }

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (x[i] < x[j]) {
                int temp = x[i];
                x[i] = x[j];
                x[j] = temp;
            }
        }
    }
}
```

```
    }  
}  
  
printf("The second minimum element is %d\n", x[n - 2]);  
printf("The second maximum element is %d\n", x[1]);  
  
return 0;  
}
```

Output:



```
Enter number of elements: 6  
Enter the elements: 17  
4  
5  
6  
2  
10  
The second minimum element is 4  
The second maximum element is 10  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Exercise –3

Code:

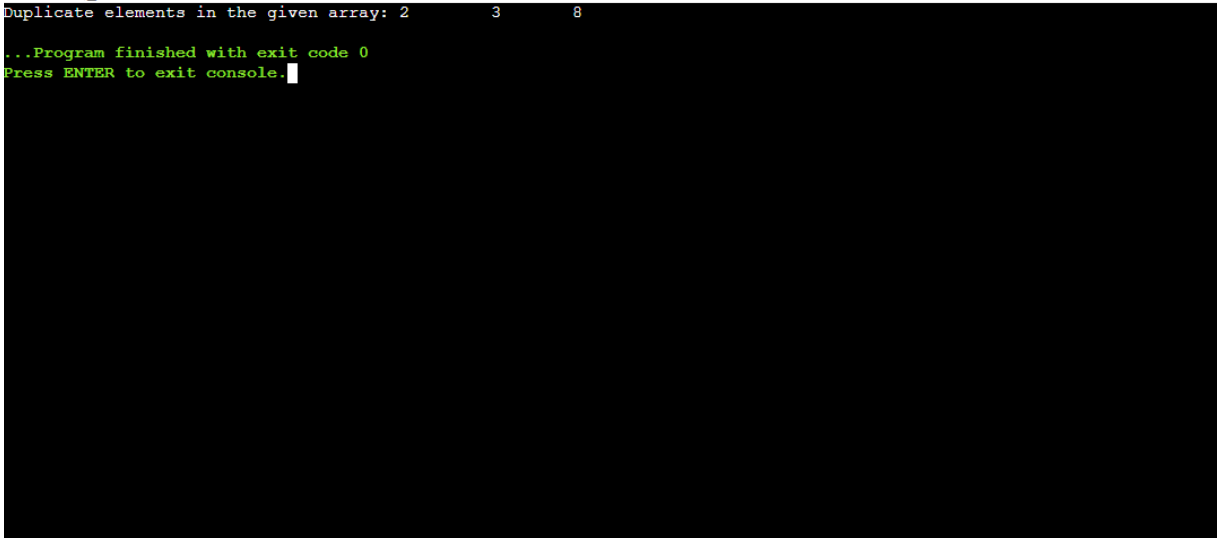
```
#include <stdio.h>
```

```
int main() {  
    int arr[] = {1, 2, 3, 4, 2, 7, 8, 8, 3};  
    int n, i, j, length;  
    length = sizeof(arr) / sizeof(arr[0]);  
    int found[length];  
  
    printf("Duplicate elements in the given array: ");  
  
    for (i = 0; i < length; i++) {  
        found[i] = 0;  
    }  
  
    for (i = 0; i < length; i++) {  
        if (found[i] == 1) {  
            continue;  
        }  
  
        for (j = i + 1; j < length; j++) {  
            if (arr[i] == arr[j]) {  
                found[j] = 1;  
                printf("%d\t", arr[j]);  
            }  
        }  
    }  
}
```

```
}

return 0;
}
```

Output:

A screenshot of a terminal window with a black background and green text. The text shows the output of a program: 'Duplicate elements in the given array: 2 3 8', followed by '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor at the end.

```
Duplicate elements in the given array: 2 3 8
...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise – 4

Code:

```
#include <stdio.h>

int main() {
    int x[50], i, n, p;
```

```
printf("Enter the number of elements: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0 || n > 50) {
```

```
    printf("Invalid number of elements. Please enter a value between 1  
and 50.\n");
```

```
    return 1;
```

```
}
```

```
printf("Input array elements:\n");
```

```
for (i = 0; i < n; i++) {
```

```
    scanf("%d", &x[i]);
```

```
}
```

```
printf("Input position to delete: ");
```

```
scanf("%d", &p);
```

```
if (p < 0 || p >= n) {
```

```
    printf("Invalid position to delete. Please enter a value between 0  
and %d.\n", n - 1);
```

```
    return 1;
```

```
}
```

```
    for (i = p; i < n - 1; i++) {  
        x[i] = x[i + 1];  
    }  
  
    n = n - 1;  
  
    printf("Array elements: ");  
  
    for (i = 0; i < n; i++) {  
        printf("%d\t", x[i]);  
    }  
  
    return 0;  
}
```

Output:

```
Enter the number of elements: 5  
Input array elements:  
10  
20  
30  
40  
50  
Input position to delete: 1  
Array elements: 10    30    40    50  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Exercise –5

Code:

```
#include <stdio.h>

void insertElement(int arr[], int n, int num, int p) {
    for (int i = n; i >= p; i--) {
        arr[i] = arr[i - 1];
    }
    arr[p - 1] = num;
}

int main() {
    int x[50], n, num, p;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Input array elements:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &x[i]);
    }
}
```

```
printf("Input element to insert: ");
```

```
scanf("%d", &num);
```

```
printf("Input position where to insert: ");
```

```
scanf("%d", &p);
```

```
insertElement(x, n, num, p);
```

```
n++;
```

```
printf("Array elements after insertion: ");
```

```
for (int i = 0; i < n; i++) {
```

```
    printf("%d\t", x[i]);
```

```
}
```

```
return 0;
```

```
}
```

Output:

```
Enter the number of elements: 5
Input array elements:
10
20
30
40
50
Input element to insert: 25
Input position where to insert: 3
Array elements after insertion: 10    20    25    30    40    50
...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise –6

Code:

```
#include <stdio.h>

int main() {
    int x[50], n, i;

    printf("Enter no of elements: ");
    scanf("%d", &n);
    printf("Enter the element: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &x[i]);
    }

    int isPalindrome = 1;

    for (i = 0; i < n / 2; i++) {
```

```
    if (x[i] != x[n - i - 1]) {  
        isPalindrome = 0;  
        break;  
    }  
}  
  
if (isPalindrome) {  
    printf("The array is palindrome");  
} else {  
    printf("The array is not palindrome");  
}  
  
return 0;  
}
```

Output:

```
Enter no of elements: 5
Enter the element: 1
2
3
2
1
The array is palindrome

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise –7

Code:

```
#include <stdio.h>

int main() {
    int x[50], reverse[50], i, n;
    printf("Enter no of elements:");
    scanf("%d", &n);
    printf("Input array elements:\n");

    for (i = 0; i < n; i++) {
        scanf("%d", &x[i]);
    }
}
```

```
for (i = 0; i < n; i++) {  
    reverse[i] = x[n - 1 - i];  
}  
  
printf("Original Array:");  
for (i = 0; i < n; i++) {  
    printf("%d", x[i]);  
}  
  
printf("\nThe Reverse array:");  
for (i = 0; i < n; i++) {  
    printf("%d", reverse[i]);  
}  
  
return 0;  
}
```

Output:

```
Enter no of elements:5
Input array elements:
1
2
3
4
5
Original Array:12345
The Reverse array:54321

...Program finished with exit code 0
Press ENTER to exit console.
```